Piyush Rajendra Nagpal

Linkedin: linkedin.com/piyush-nagpal
Github: github.com/its-me-piyush
Email: piyushnagpal777@gmail.com
Mobile: (720) 755-2088

Education

- **University of Colorado Boulder** — Boulder, USA
  Professional Master's in Embedded Systems Design; GPA: 3.77/4 — *2025 - 2027*
  Courses: *Embedded Software Systems, PCB Design, Embedded AI, IoT*
- **International Institute of Information Technology (I2IT)** — Hinjiwadi, India
  Bachelor of Technology - Electronics and Telecommunication; GPA: 8.7/10 — *2019 - 2023*

Technical Skills

- **Programming**: C, C++, Python, Bash, Dart, Java
- **Embedded Systems**: STM32, ESP32, Raspberry Pi, Arduino, bare-metal firmware, embedded Linux, RTOS concepts
- **Interfaces/Protocols**: UART, SPI, I2C, CAN, BLE, Wi-Fi, MQTT, gRPC, Protocol Buffers
- **Hardware/Debugging**: PCB design, Altium Designer, oscilloscopes, logic analyzers, ADC/DAC, timers, DMA, power integrity
- **Tools**: Git, Docker, Linux, Makefiles, VS Code, secure coding practices

Experience

- **Qualcomm** — *Boulder, CO*
  Interim Engineering Intern – Infrastructure Test Team — *May 2026 – Aug 2026*
  - Validate the Linux release of an internal test-tracking platform for compatibility with Linux-based 5G infrastructure test environments.
  - Migrate and modify Python infrastructure test scripts to improve execution visibility, logging, test tracking, and result analysis.
  - Diagnose integration issues across Python scripts, Linux environments, and internal testing workflows in collaboration with the 5G infrastructure test team.
- **Cognologix Technologies** — *Pune, India*
  Product Developer — *July 2023 – July 2025*
  - Developed embedded firmware and IoT product features for BLE-enabled systems, integrating microcontroller firmware, Flutter mobile applications, backend services, and hardware interfaces.
  - Implemented and debugged real-time communication over BLE, MQTT, and CAN, reducing field failures by 40% and improving device-to-cloud reliability.
  - Built and validated UART, SPI, I2C, GPIO, timer, and sensor-data interfaces across production IoT deployments supporting more than 10,000 active devices.

Projects

- **Instrument Droid – 4-Layer Electronic Load System**: Designed a 4-layer microcontroller-based electronic load to characterize VRM Thevenin voltage and resistance. Implemented DAC-controlled MOSFET current sinking, multi-channel ADC sampling, UART reporting, PCB layout, hardware bring-up, and calibration. Used oscilloscopes and measurement tools to validate power behavior, signal integrity, and system performance.
- **Golden Arduino – Custom Arduino-Compatible Board**: Designed and built an ATmega328-based Arduino-compatible board with CH340G USB-UART, crystal oscillator, voltage regulation, ESD protection, and optimized decoupling. Completed schematic capture, PCB layout, bootloader flashing, and hardware bring-up. Validated serial communication, switching noise, and power stability using oscilloscopes and logic analyzers.
- **AESD Rover Autonomy System – Yocto, ESP32, CAN**: Built a distributed embedded rover with Raspberry Pi Yocto Linux high-level control and ESP32 real-time motor/sensor control. Implemented CAN communication using SocketCAN/MCP2515, structured drive, telemetry, heartbeat, and emergency-stop messages. Integrated dual L298N motor drivers, ultrasonic obstacle detection, and watchdog-based fail-safe behavior for manual and autonomous control.
- **BLE-MQTT-CAN RV Awning Control System**: Developed an embedded Linux-based RV awning control prototype using Raspberry Pi as a BLE peripheral and MQTT gateway. Implemented BLE services for mobile app communication, MQTT messaging for smart hub integration, and CAN commands for motor extend, retract, and stop control. Designed telemetry flow for motor current, encoder count, latch state, wind speed, tilt, and board status.

Honors

- Patron Award for Overall Academic Achievement; Copyright Registration ROC: L-123685/2023; Finalist, Elastic Run #Possibilities Hackathon; Finalist, Rakathon by Rakuten